INCH-POUND

MIL-R-44471
30 July 1993

MILITARY SPECIFICATION

RANCHERO SAUCE, FREEZE-DEHYDRATED, (OPERATIONAL RATION COMPONENT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers freeze-dehydrated, ranchero sauce for use by the Department of Defense as a component of operational rations (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	-	Aluminum Foil
PPP-B-26	-	Bag, Plastic (General Purpose)
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5018, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8950

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PPP-C-1752 - Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible
PPP-C-1797 - Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam
PPP-T-60 - Tape: Packaging, Waterproof

MILITARY

MIL-E-44470 - Eggs, Scrambled, Cooked, Dehydrated, (Operational Ration Component)

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Tomato Paste

U.S. Standards for Grades of Canned Tomatoes

(Copies are available from the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0709, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

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U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, One Maritime Plaza, 23rd Floor, San Francisco, CA 94111.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by Density Gradient Technique

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

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NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

THE UNITED STATES PHARMACOPEIA CONVENTION, INC.

The United States Pharmacopeia (USP) and the National Formulary (NF)

(Application for copies should be addressed to the United States Pharmaceutical Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Tomatoes, diced, canned. The canned, diced tomatoes shall be U.S. Grade B or better of the U.S. Standards for Grades of Canned Tomatoes. The canned, diced tomatoes shall have a tomato solids content of not less than 6 percent and shall be of the latest season's crop (see 6.5).

3.2.2 Water. Water used for formulation, blanching, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.3 Tomato paste. Tomato paste shall be U.S. Grade A of the U.S. Standards for Grades of Tomato Paste. The tomato paste shall have a tomato solids content of not less than 26 percent and shall be of the latest season's crop.

3.2.4 Jalapeno peppers, sliced, canned. The canned jalapeno peppers shall be 1/4 inch thick slices. The jalapeno peppers shall be packed in cans or jars. The jalapeno peppers shall be of the latest season's pack.

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3.2.5 Onions, dehydrated, chopped. Dehydrated chopped onions shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.6 Oil, vegetable. Vegetable oil shall possess a clean, bland flavor, and shall have a minimum stability of 25 hours (A.O.M.), have a free fatty acid value not to exceed 0.05 percent, have a moisture/volatile matter content not to exceed 0.06 percent, and shall pass a cold test of 5.5 hours (minimum). The oil may contain antioxidants in combinations and amounts permitted by FDA regulations.

3.2.7 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.8 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or the outer and inner covering have been removed. The ground pepper shall have a characteristic, penetrating odor, a hot, biting, pungent flavor, and a light color. The ground pepper shall contain not less than 1.0 mL of volatile oil per 100 grams of ground white pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.9 Oregano, ground. Ground oregano shall be derived from the dried leaves of Origanum vulgare L. and shall possess a strong camphoraceous aroma and a pungent, slightly bitter flavor. The ground oregano shall contain not less than 2.0 mL of volatile oil per 100 grams of ground oregano and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.

3.2.10 Cumin, ground. Cumin shall be the ground, true aromatic substance derived from Cuminum cyminum L. from which no volatile or other flavoring constituents have been removed, and shall be free from artificial coloring and impurities. The ground cumin shall contain not less than 2.2 mL of volatile oil per 100 grams of ground cumin and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.

3.2.11 Garlic powder. Garlic powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.12 Chili powder. Chili powder shall be a blend of 77 to 82 percent chili pepper; not less than 8.0 percent cumin; not less than 4.0 percent oregano; not less than 1.0 percent garlic; and not more than 8.0 percent salt. Anticaking agents may be included at a level not to exceed 2.0 percent. The chili powder shall have a Scoville Pungency Value of not less than 900 units and the moisture content shall not exceed 13.0 percent. Not less than 95 percent, by weight, of the chili powder shall pass through a U.S. Standard No. 20 Sieve.

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3.2.13 Nitrogen. Nitrogen shall be U.S. Pharmacopoeia grade and shall be water or liquid nitrogen pumped.

3.2.14 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formulation.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Preparation of ranchero sauce. The ranchero sauce shall be formulated and prepared as follows:

Ingredients	Percent by weight
Tomatoes, diced, canned (6% solids) 1/	71.00
Water	14.35
Tomato, paste (26% solids) 1/	5.40
Jalapeno peppers, sliced, canned, drained	3.50
Onions, dehydrated, chopped	2.67
Oil, vegetable	2.33
Salt	0.49
Pepper, white, ground	0.09
Oregano, ground	0.08
Cumin, ground	0.05
Garlic powder	0.02
Chili powder	0.02

1/ If the tomato solids are not available, the tomato solids that are used shall be adjusted, as necessary, to bring the solids level to that specified.

The product shall be prepared as follows:

a. Heat water, dehydrated onions, and spices to 180°F.

b. Add the diced tomatoes, tomato paste, oil and drained jalapeno peppers and heat to 160°F.

c. The product shall be frozen and freeze-dehydrated in accordance with 3.4 and 3.4.1.

d. The freeze-dehydrated product shall be pressed through a 1/2 inch mesh sieve or screen.

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3.4 Freezing. The prepared product shall be spread on dehydrator trays and cooled immediately, placed in a freezer within 4 hours after cooking, and frozen to 0°F or below within 8 hours after placing in the freezer. The frozen product shall be maintained at 0°F or below at all times prior to freeze dehydration, and shall be in the freeze dehydration process not less than 5 days after freezing.

3.4.1 Dehydration. The product shall be freeze-dehydrated at an absolute pressure not to exceed 1.5 mm of mercury for more than 5 minutes during the first three quarters or 10 minutes during the last quarter of the drying cycle. At no time shall there be evidence of thawing in the product. The product temperature shall not exceed 150°F. If the platen temperature is maintained at 155°F or below with radiant heating, the product temperature may be disregarded. After dehydration is completed, the pressure shall be equalized to atmospheric level with nitrogen. In no case shall more than 16 hours elapse between the time the chamber is opened to the time the product is completely packaged. During this interim period, the product shall be adequately protected from oxygen and moisture by either holding under a nitrogen atmosphere with 2.0 percent or less oxygen, or under a vacuum of at least 27 inches of mercury for the entire period. If vacuum is used, it shall be broken with nitrogen. Alternate product holding procedures may be used (see 6.6).

3.5 Pouch filling and sealing. The freeze-dehydrated product shall be packaged immediately in accordance with 5.1.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

Dehydrated product:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. There shall be no evidence of incomplete dehydration such as wet or soft spots.

e. There shall be no evidence of faulty dehydration such as glazed surface areas measuring more than 0.5 inch in any dimension.

f. There shall be no evidence of faulty dehydration such as burnt areas measuring more than 0.5 inch in any dimension.

g. There shall be no evidence of faulty dehydration such as dark colored areas of any dimension.

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h. The moisture content of the product shall be not greater than 2.0 percent.

i. The net weight of the product shall be not less than that specified below:

Single service (grams)	Multi-service (grams)
35	100

j. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample.

k. The E. coli count shall be less than 3 per gram (no positive tubes in the standard three tube MPN technique) in four of five samples and not greater than 10 per gram in any sample.

Rehydrated product:

l. When the product is rehydrated (see 4.5.6), the product shall show complete water penetration.

m. The rehydrated product shall possess the odor, flavor, and texture considered normal for freeze-dehydrated ranchero sauce. There shall be no scorched, burnt, or foreign flavor or odor.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.2) in palatability and overall appearance.

3.7 Plant qualifications. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110: "Current Good Manufacturing Practice in Manufacturing, Packing or Holding Human Food," and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

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4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.2), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the AOAC International, and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.1.2 Material certification. A certificate of compliance may be accepted as evidence that the characteristics listed below conform to the specified requirements.

Material requirement	Requirement paragraph	Test procedure
Mixing bag.		
Material identification	5.1.2	Laboratory evaluation

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Polyethylene film thickness	5.1.2	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Color of material	5.1.2	Visual evaluation
Baseboard (when used).		
Material identification	5.1.2.2	Laboratory evaluation
Paperboard film thickness	5.1.2.2	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Color of material	5.1.2.2	Visual evaluation
Pouch.		
Laminated material identification and construction	5.1.3 and 5.1.3.1	Laboratory evaluation.
Polyethylene film and polyester film thickness	5.1.3	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Aluminum foil thickness	5.1.3	As specified in QQ-A-1876
Color of laminated material	5.1.3	Visual evaluation

4.5.1.3 Unfilled mixing bag examination. The unfilled mixing bags shall be examined for the defects listed in table I. The lot size shall be expressed in bags. The sample unit shall be one mixing bag. The inspection level shall be S-4 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65.

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TABLE I. Unfilled mixing bag defects

Category	Defect
Major	
101	Not formed as specified
102	Inside dimensions not as specified
103	Tear, hole, or open seal
104	Not material specified
105	Bags stick together
106	Not clean
107	Foreign odor

4.5.1.4 Unfilled mixing bag leakage testing. The unfilled mixing bag shall be tested for leakage as follows: Each sample of 5-1/2 (length) by 3-1/2 (depth) by 8 (height) inch size mixing bag shall be filled with 300 mL of boiling water. Each sample of 9 (length) by 3-1/2 (depth) by 12 (height) inch size mixing bag shall be filled with 1000 mL of boiling water. Care shall be taken to prevent spillage of water onto the outside of the bag. Each water filled bag shall be suspended 8 to 12 inches above a sheet of clean paper for 10 minutes. Any moisture on the paper attributable to bag leakage shall be classified as a major defect. The lot size shall be expressed in bags. The sample unit shall be one unfilled mixing bag. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation, or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch examination. The filled and sealed pouches shall be examined for the defects listed in table II. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 4.0 for minor defects.

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TABLE II. Filled and sealed pouch defects 1/

Category		Defect
Major	Minor	
101		Tear, hole, or abrasion through one or more layers 2/
102		Not heat sealed as specified
103		Presence of foldover wrinkles and entrapped product that reduces the effective closure seal to less than 1/16 inch 3/
104		Evidence of delamination 4/
105		Not clean 5/
106		Required labeling or marking missing, incorrect, illegible, or that smudges
107		Any impression or design on the heat seal surface which conceals or impairs visual detection of seal defects 6/
108		Closure seal widths not as specified
	201	Evidence of delamination 4/
	202	Color of pouch not as specified
	203	Tear notch missing
	204	Tear notch not located as specified
	205	Depth of tear notch not as specified
	206	Serrations missing 7/
	207	Serrations not located as specified 7/
	208	Depth of serrations not as specified 7/
	209	Closure or top seal extends into or below tear notch

1/ Any evidence of rodent or insect infestation shall be cause for rejection of the lot.

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2/ Pinholes or breaks inherent to the manufacturing process of the aluminum foil shall not be scored as a defect.

3/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/16 inch wide from side seal to side seal that produces a hermetically sealed pouch.

4/ Delamination defect classification:

Major - Delamination of the outer ply in the pouch seal area that can be propagated to expose aluminum foil at the food product edge of the pouch after manual flexing of the delaminated area. To flex, the delaminated area shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delamination area shall then be rapidly flexed 10 times by rotating both hands in alternating clockwise - counterclockwise directions. Care shall be exercised when flexing delaminated area near the tear notch to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal or if the separated area is too small to be held between thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to the product edge of the seal with no discernible resistance to the gentle lifting, the delamination shall be scored as a major defect. Additionally, spot delamination of the outer ply in the body of the pouch that is able to be propagated beyond its initial borders is also a major defect. To determine if the delaminated area is a defect, use the following procedure: Mark the outside edges of the delaminated area using a bold permanent marking pen. Open the pouch and remove the contents. Cut the pouch transversely not closer than 1/4 inch (± 1/16 inch) from the delaminated area. The pouch shall be flexed in the are area in question using the procedure described above. Any propagation of the delaminated area, as evidenced by the delaminated area exceeding the limits of the outlined borders, shall be scored as a major defect.

Minor - Minor delamination of the outer ply in the pouch seal area is acceptable and shall not be scored as a minor defect unless it extends to within 1/16 inch of the food product edge of the seal. All other minor outer ply delamination in the pouch seal area or isolated spots of delamination in the body of the pouch that do not propagate when flexed as described above shall be scored as a minor defect.

5/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal fillings) or generally detracts from the clean appearance of the pouch. The following examples shall not be scored as defects for unclean:

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a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the pouch or by gently brushing the pouch with a clean dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

6/ If doubt exists as to whether or not the sealing equipment leaves an impression or design on the closure seal surface that could conceal or impair visual detection of seal defects, samples shall be furnished to the contracting officer for a determination as to acceptability.

7/ Applicable only to pouches made on machinery that produces serrations.

4.5.4 Pouch vacuum examination. Not less than 96 hours after filling and sealing, the pouches shall be visually examined for conformance to the vacuum requirements in 5.2. The sealed pouch shall continue to exhibit tight adherence to the surface contours of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between thumb and forefinger of each hand, simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.5 Net weight examination. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty mixing bag and pouch. Any individual net weight of less than that specified in 3.6 shall be classified as a minor defect. The results shall be reported to the nearest gram. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed mixing bag. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5. In addition, the lot shall be rejected if the sample average net weight is less than that specified in 3.6.

4.5.6 Product examination. The finished product shall be examined for the defects listed in table III. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

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TABLE III. Product defects 1/ 2/

Category	Defect
Major	
101	Pouch or mixing bag has foreign odor
	Dehydrated product
102	Presence of incomplete dehydration such as wet or soft spots
103	Presence of glazed surface areas measuring more than 0.5 inch in any dimension
104	Presence of burnt areas measuring more than 0.5 inch in any dimension
105	Presence of dark colored cores in any dimension
	Rehydrated product 3/
106	Product shows incomplete water penetration
107	The rehydrated product does not possess the odor, flavor, or texture considered normal for freeze-dehydrated ranchero sauce

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, stale, sour, rancid, moldy), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.6.1).

3/ The product shall be rehydrated as follows:

a. Single service pouch - Add 300 mL of hot water (180°F to 200°F) to mixing bag and allow to rehydrate for 10 minutes prior to examination.

b. Multi-service pouch - Add 1000 mL of hot water (180°F to 200°F) to mixing bag and allow to rehydrate for 10 minutes prior to examination.

4.5.7 Internal pressure test. The pouches shall be emptied prior to testing (remove dehydrated product and mixing bag) for use with three-seal tester only. Internal pressure resistance shall be determined by

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pressurizing the pouches while they are restrained between two rigid plates spaced 2-3/4 inches (± 1/8 inch) apart. If a three-seal tester (one that pressurizes the pouch through an open end) is used, the closure seal shall be cut off for testing the side and bottom seals of the pouch; for testing of the closure seal, the bottom seal shall be cut off. If a four-seal tester (designed to pressurize filled pouches by use of a hypodermic needle through the pouch wall) is used, all four seals can be tested simultaneously. Pressure shall be applied at the approximate uniform rate of 1 pound per square inch gage (psig) per second until 14 psig pressure is reached. The 14 psig pressure shall be held constant for 30 seconds and then released. The pouches shall then be examined for separation or yield of the heat seals. Any rupture or evidence of seal separation greater than 1/16 or seal separation that reduces the effective closure seal width to less than 1/16 inch shall be considered a test failure. The lot size shall be expressed in pouches. The sample unit shall be one pouch. The inspection level shall be S-1. Any test failure shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.8 Moisture content testing. Eight filled and sealed pouches shall be selected at random from the lot regardless of lot size. The product shall be individually tested for moisture content in accordance with the Official Methods of Analysis of AOAC, method 925.45, using a sample containing about 2 grams at a temperature of 70°C for 16 hours at a pressure of 100 mm of mercury. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the moisture content requirement in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.9 Microbiological testing. Five filled and sealed pouches of finished product shall be selected at random from the lot regardless of lot size. The product shall be individually tested for microbiological levels in accordance with the Official Methods of Analysis of AOAC, methods 966.23 and 966.24. The diluent shall be added to each sample and allowed to stand for 15 minutes before blending the sample. Any result not conforming to the microbiological requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.10 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: Markings missing, incorrect, or illegible.
Cushioning material missing or not as specified.

Minor: More than 35 pounds packed in a box.
Bags not packed flat in layers.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

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5.1.1 Level A preservation. The product shall be filled into the mixing bag as specified in 5.1.2 and sealed in pouches as specified in 5.1.3.

5.1.2 Mixing bag. The mixing bag shall be fabricated from 0.003 inch thick polyethylene made in accordance with type I bags of PPP-B-26, except that the bag shall be made from food grade polyethylene. The mixing bag shall be approved by the FDA for food use and shall not impart any undesirable odor or flavor to the product. The bottom seal shall maintain its strength and integrity when tested as specified in 4.5.1.4. The inside dimensions of the single service and multi-service bag shall be 5-1/2 inches (± 1/8 inch) in length by 3-1/2 inches (± 1/8 inch) in depth by 8 inches (± 1/8 inch) in height, with the 5-1/2 inch dimension being the opening of the bag. Alternatively, for single service bags, polyethylene bags conforming to type II of PPP-B-26, with inside dimensions of 9 inches (± 1/8 inch) in width by 8 inches (± 1/8 inch) in length, with the 9 inch dimension being the opening of the bag with the open end of the bag may be used. When type II bags are used, they shall be formed over a baseboard on a mandrel as specified in 5.1.2.1 before filling. While the bag is positioned on the mandrel, the two bag ears shall be taped down over a 5-1/2 by 3-1/2 inch baseboard constructed in accordance with 5.1.2.2. An 8 inch long strip of 1 inch wide tape complying with type III, class 2 of PPP-T-60 shall be applied across both ears to fasten them securely to the baseboard. Alternatively, both type I and type II mixing bags may be fabricated from unpigmented 0.003 inch thick high density 0.941 to 0.965 g/cm food grade, flat polyethylene tubing. The bottom seal of type I and type II mixing bags fabricated from the alternate material shall maintain its strength and integrity when tested as specified in 4.5.1.4.

5.1.2.1 Mandrel. The mandrel face shall be 5-3/8 inches in length by 3-3/8 inches in width and the height shall not be less than 12 inches. All corners and edges of the mandrel shall be rounded and all surfaces finished to prevent damage to the bag material.

5.1.2.2 Baseboard. The baseboard shall be constructed from unbleached and uncoated solid sulfate or sulfite paperboard not less than 0.022 inch thick. Alternatively, the baseboard may be constructed from 0.022 inch thick food grade paperboard made from recycled fibers. If paperboard made from recycled fibers is used, it shall be natural color ranging from brown to grey. The baseboard shall be 5-1/2 inches (± 1/8 inch) in length by 3-1/2 inches (± 1/8 inch) in width and shall have all corners rounded to form a smooth contour.

5.1.3 Pouch material. The pouch shall be fabricated from a 3-ply flexible laminate material consisting of an outer ply of 0.0005 inch (± 0.0001 inch) thick polyester bonded to an intermediate ply of 0.0007 inch (± 0.00014 inch) thick aluminum foil with 10 pounds per ream pigmented polyethylene and bonding the opposite side of the aluminum foil to 0.003 inch (± 0.0006 inch) thick ionomer (inner ply) or a blend of not less than 50 percent linear low density polyethylene and polyethylene. The linear low density polyethylene portion of the blend shall be the copolymer of ethylene and octene-1, having

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a melt index range of 0.8 to 1.2 grams per 10 minutes in accordance with ASTM D 1238 and a density range of 0.918 to 0.922 g/cm in accordance with ASTM D 1505. The complete exterior of the pouch shall be uniformly colored in the range of 34079 through 34087, or in the range of 24052 through 24087, or in the range of 30045 through 30118, excluding color 30109 of FED-STD-595. The color application process shall not adversely affect the material performance. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being packaged. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches.

5.1.3.1 Pouch construction. The preformed pouch shall be a flat style pouch having inside dimensions of 6 inches (± 1/8 inch) in width by 9 inches (± 1/8 inch) in length for the single service pouch, with the 6 inch dimension being the opening of the pouch between the heat sealed sides. The pouch shall be made by heat sealing three edges (two sides and bottom) with 3/8 inch (± 1/8 inch) wide seals. The pouch shall withstand an internal pressure of 14 psig for 30 seconds without rupture or seal separation greater than 1/16 inch or seal separation that reduces the effective closure seal width to less than 1/16 inch when tested as specified in 4.5.7. A V-shaped, C-shaped or U-shaped tear notch at least 1/32 inch deep, located 1 to 1-1/4 inches from the top edge of the pouch (excluding the lip) shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be not less than 3/16 inch. One side of the open end of the pouch may be provided with an extended or foldover lip, extended not more than 5/16 inch (± 1/16 inch) to facilitate opening and filling. Tear notch location shall be measured from the top of the pouch, excluding the extended or foldover lip. Alternatively, if the pouch has serrated edges, the serrations may be used as tear notches provided that the serrations are sharp (no plastic tailing exists) and the serration depth and the minimum seal width at the serrated edges are in accordance with the above tear notch requirements. The pouch shall be constructed with square or rounded corners at all four corners (radius approximately 3/8 inch).

5.2 Filling and sealing. The product shall be filled into the mixing bag as specified in 5.1.2 in accordance with the net weight requirements as specified in 3.6. After filling, the end panel of the bag (corresponding to the 3-1/2 inch dimension) shall be folded in, and the bag shall be closed by folding over its open end. Where required by the bulkiness of the product, the bulk of the contents within the bag shall be compressed such that the product thickness is not more than 2-1/4 inches by applying a constant uniform pressure over the entire top surface of the product. The filled bag with the open end folded over to prevent spillage of the product shall be inserted into the bottom of the pouch as specified in 5.1.3 in such a manner to avoid puncturing the pouch material or contaminating the heat seal area. The filled pouches shall be sealed under a vacuum level of 23 inches of mercury, and applying a continuous heat seal. The filled pouches shall have a 3/8 inch (± 1/8 inch) heat seal. If thermal impulse closure seals are used, a seal width of 1/8 inch will be acceptable. The closure seal shall be

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free of foldover wrinkles, delamination, or entrapped product that reduces the effective closure seal to less than 1/16 inch wide. The filled and sealed pouches shall withstand an internal pressure of 14 psig for 30 seconds without rupture or seal separation greater than 1/16 inch or seal separation that reduces the effective closure seal width to less than 1/16 inch, when tested as specified in 4.5.7. The filled and sealed pouch shall show no loss of vacuum when examined as specified in 4.5.4.

5.3 Packing for shipment to ration assembler. Not more than 35 pounds of product shall be packed flat in a fiberboard box constructed and closed in accordance with style RSC-L, class domestic, variety SW, grade 200 of PPP-B-636. Cushioning material not less than 1/8 inch thick, conforming to type II class B of PPP-B-843 or cushioning material not less than 1/6 inch thick, conforming to PPP-C-1797 or cushioning material conforming to type 7, class 1 of PPP-C-1752 shall be placed between each of the layers and in the bottom and top of the box to minimize movement of the individual packages.

5.4 Labeling and marking.

5.4.1 Pouch. Each pouch shall be clearly printed or stamped, in a manner that does not damage the pouch, with permanent black ink which is free of carcinogenic elements or ingredients. The information shall be located on the body of the pouch not closer than 1/16 inch to any seal. All pouches shall be coded. If a non-contact type printer is used, the information may be located anywhere on the pouch (in one complete print), except in the closure seal area. The label must include the following information:

Product name (letters not less than 7/32 inch high)
Rehydration instructions (letters not less than 3/32 inch high)
Ingredients (letters not less than 3/32 inch high)
Date of production/lot number 1/ (letters not less than 1/8 inch high)
Net weight (letters not less than 1/8 inch high)
Official establishment number (for example, EST 38) or a three letter code identifying the establishment (letters not less than 1/8 inch high)
Name and address of producer (letters not less than 1/8 inch high)

1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1993 would be coded as 3078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code.

5.4.1.1 Rehydration instructions. Each pouch label shall display the rehydration instructions as follows:

a. For single service pouches (letters not less than 1/8 inch high):

To rehydrate add 6 ounces water, stir, wait 2 to 5 minutes.

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b. For multi-service pouches (letters not less than 1/8 inch high):

To rehydrate add 15 to 16 ounces water, stir, wait 3 to 5 minutes.

5.4.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The ranchero sauce is intended to be served on scrambled egg products or eaten with other components as a condiment. Scrambled egg product requirements are specified in MIL-E-44470.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
c. When a first article is required (see 3.1, 4.4, and 6.3).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.5 Tomatoes, diced, canned. It has been found that canned, diced tomatoes manufactured by Pacific Coast Producers, Stockton, CA, meets the requirements of 3.2.1 and performed satisfactory in this product. Other equivalent canned, diced tomatoes may be used.

6.6 Alternate product holding procedures. The contracting officer may authorize alternate contractor-recommended product holding procedures if the the alternate procedures are approved by the U.S. Army Natick, Research, Development and Engineering Center.

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6.7 Subject term (key word) listing.

Combat field feeding
Food processing
Freeze-dehydrated food
Salsa

Custodians:

Army - GL
Navy - MC

Preparing activity:

Army - GL

(Project 8950-0342)

Review activities:

Army - MD, QM
Navy - SA
Air Force - 50
DLA - SS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-44471	1993 July 30

3. DOCUMENT TITLE

RANCHERO SAUCE, FREEZE-DEHYDRATED, (OPERATIONAL RATION COMPONENT)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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